

RESEARCH PROPOSAL

Surrogate-Aware Runtime Assurance for Agentic Medical LLMs

Preventing evidence overclaiming in clinician-facing AI systems

Author: Ravi Bajaj | **Target venue:** ML4H 2026 Proceedings Track | **Course:** AI in Healthcare

Problem and Hypothesis

Medical LLMs are often tested for hallucination, but agentic workflows create a subtler failure: a real citation, workflow metric, surrogate endpoint, population-mismatched result, or privacy-limited genetic release can be converted into a stronger patient-outcome claim than the evidence supports. The practical target is recurring hospital genetics: pharmacogenomic alerts, newborn-screening follow-up, ACMG secondary findings, hereditary cancer/FH/Lynch flags, VUS handling, and family-linked privacy boundaries. Hypothesis: a runtime assurance monitor can reduce unsafe clinical overclaiming by gating each LLM output or proposed action by citation validity, endpoint strength, study design, population fit, context, opportunity cost, clinician authority, privacy budget, and consent boundary.

Key Background

- Stegenga, Medical Nihilism: motivates caution when evidentiary pipelines overstate treatment benefit.
- Byrne, Domenico, and Moore: pragmatic patient-level RCTs remain the standard for AI patient-outcome claims.
- FDA-NIH BEST / FDA surrogate endpoint materials: distinguish clinical outcomes, validated surrogates, ordinary surrogates, and biomarkers.
- CPIC, ACMG, HRSA/RUSP, ACT sheets, and CDC Tier 1 materials anchor the project in hospital-facing genetics rather than speculative genetics.
- Dwork and Roth: differential privacy supplies measurable privacy loss, composition, post-processing, and group/family privacy concepts.
- Loh-style statistical genetics, Arrow social choice, Harchol-Balter scheduling, LDA, and Mahalanobis distance motivate population-structure, consent-aggregation, queueing, mixture-model, and covariate-transport gates.
- Hou, Ji, Zhang, and Stefanidis on Kolmogorov-Arnold Networks: architecture/theorem claims can guide model selection, but remain method evidence rather than clinical permission.

Method and Stage 1 Results

Stage 1 is implemented as a deterministic synthetic assurance harness with 30 scenarios and 30 matched runtime decisions. The controller may allow bounded second-reader use, narrow to surrogate/process language, allow simulation-only rehearsal, require provenance/proof-status or consent audit, stress-test generalization/order/confounding, or deny the claim. A ZDD audit compactly represents admissible sparse evidence states; a model-selection layer chooses among 19 claim models; all code and JSON/CSV results are included.

- ZDD/model selection/proof status: 23 observed subsets over 39 features; 30/30 actions matched; the KAN architecture case was narrowed to method evidence; the proof-status poset allowed hard-outcome permission in 4/4096 states and kept NIETTU audit-only.
- Runtime resources: 24 orderings tested; heterogeneous resources showed context reversal. Kelly-style exposure capped claim authority after all-in ruin probability 1.000.
- Confounding and witnesses: orthogonal projection reduced max design inner product to $4.68e-15$; biased strongest-five witness selection had variation distance 0.828.
- Hospital genetics: the harness includes recurring genetics-result triage; family preferences produced a cycle, full raw release was blocked, and partial inclusion stayed runtime-safe with composed epsilon 0.85.
- Population/source/boundary transport: event algebra verified measurable identities; measure-on-measures max TV was 0.106; Mahalanobis flagged genomeindia_like, latin_america_admixed_like; Hex scaling sampled 36000 boards with 0 ambiguous terminals and found generic smoothing could flip the boundary at rate 0.256.
- Multimodal and universality transport: picture-language audit classified 9 diagrammatic artifacts with zero hard-outcome permissions; loop-equation stability kept the stable transfer chain at error 0.0173 under budget 0.050; switching-cumulant checks canceled the main quadratic term and kept rare-event/replacement errors inside budget; branch-factor audit allowed the separated synthetic branch but blocked the near-collision family/population branch.

Value and Stage 2

The proposal contributes an auditable standard for permissible LLM inputs and outputs: fluent text cannot upgrade evidence, and family-linked genetic features cannot be treated as ordinary single-person records. Stage 2 would apply the abstract gates to faculty-approved hospital genetics workflows first, then only later to broader polygenic-risk, wearable/telemedicine, or multilingual documentation settings. The monitor preserves clinician authority and requires patient-outcome evidence before outcome claims.

Selected References

- Stegenga (2018), Medical Nihilism. <https://global.oup.com/academic/product/medical-nihilism-9780198747048>
- Byrne, Domenico, & Moore (2024), pragmatic RCTs for AI outcomes. <https://kjonline.org/DOIx.php?id=10.3348%2Fkjr.2023.1016>
- FDA-NIH BEST Resource (2016-2025), biomarkers and endpoints. <https://www.ncbi.nlm.nih.gov/books/NBK326791/>
- Dwork & Roth (2014), Algorithmic Foundations of Differential Privacy. <https://doi.org/10.1561/04000000042>
- Bourgade & Huang (2026), Loop Equations Characterize Random Matrix Statistics. <https://arxiv.org/abs/2607.07617>